Fiscal Year:	FY 2015	**Task Last Updated:**	FY 05/27/2015
PI Name:	Dantzig, Jonathan A Ph.D.		
Project Title:	Modeling Peritectic Microstructure Formation during Directional Solidification in Space and on Earth		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	61801	**Congressional District:**	15
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2009-10 Materials Science Rapid Turn Around Flight Opportunity
Start Date:	06/03/2010	**End Date:**	06/02/2015
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 6/02/2015 per NSSC information and PI (Ed., 4/3/14) NOTE: Period of performance changed to 6/3/2010-6/2/2014 (previously 4/6/10-4/5/14) per NSSC (Ed., 7/14/10)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX10AJ73G		
Performance Goal No.:			
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Task Description:	Peritectic alloys constitute an important class of industrial alloys. The properties of these alloys depend strongly on their solidification history, and the potential exists to create novel materials with unique properties by controlling the solidification process. Peritectic alloys undergo an invariant reaction, in which a liquid phase l reacts with an existing solid phase alpha to form a second solid phase beta. When solidification occurs at very low growth rate in a high temperature gradient, this reaction can produce very complicated, non-equilibrium microstructures due to the limited diffusion in the solid state, and convection in the liquid. The European Space Agency (ESA) has an ongoing multi-national, multi-investigator program, called METCOMP, to study peritectic alloys. The proposed research will provide modeling support for one of these METCOMP projects, focusing on directional solidification in Cu-Sn alloys. Experiments performed on Earth are difficult to interpret because of the possibility of buoyancy-driven convection. Such experiments are necessary, however, to provide the scientific base for interpreting the space experiments. The objective of the proposed research is to develop and use models that do two things: (1) interact with the experimental work so as to improve the experiments themselves; and (2) provide detailed and predictive models to assist in interpretation of the experimental results. In order to accomplish the first objective, separate models for the macroscale transport phenomena and microstructure formation will be developed first. These models will then be merged into a single code that computes the transport and microstructure at the same time. This is feasible because the code will build on an existing code that uses adaptive mesh refinement to allocate computational resources efficiently.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research, if successful, will lead to improved products for peritectic alloys in general, and for Cu-Sn alloys in particular. There is experimental evidence that coupled growth in these systems can lead to novel microstructures with unique properties. However, these structures are difficult to produce and control This research could help resolve this problem, making these materials commercially available.
Task Progress:	The computational models of microstructure development in these alloys were completed in early 2014, and were included in last year's report. The main effort in this final year on this alloy system was directed to examining the origin of bubbles that form on the sides of samples solidified on Earth in prototype experiments for the upcoming ESA space experiment, which is scheduled to fly in 2016 or 2017. It was conjectured that the bubbles form as a result of the combination of poor wetting of the crucible and insufficient pressure on the melt during solidification, so that thermal and transformation shrinkage are not compensated. A redesign of one end of the crucible, in order to provide increased pressure due to capillary forces was evaluated. So far, the results are mixed, with good surface quality in one sample, and bubbles in another. This work is ongoing. It is very important for the success of the experiment that this issue be resolved, because a bubble-free surface is required to establish true one-dimensional diffusive conditions. Dendritic growth and anisotropy In earlier years of this project, computational models were used to study dendrite selection in Al-Zn dendrites. We were able to demonstrate that a systematic variation in the parameters describing the cubic anisotropy in Al-rich dendrites could explain the transition from <100> dendrites in low-Zn alloys, to <110> dendrites at high-Zn content. The intermediate region is characterized by "seaweed" structures, which we demonstrated is a product of the weak overall anisotropy due to the competition between these two mechanisms. In this past year, we extended our study to the Zn side of the Al-Zn phase diagram. These microstructures offer an unusual set of complexities, because the crystal structure of the primary phase is hexagonal, and the surface energy anisotropy is large. This leads to forbidden orientations, and dendrites with sharp edges. One of the primary determinants for dendrite orientation selection is the anisotropy of the solid-liquid surface energy. At EPFL, a new experimental technique was developed to suspend droplets in a molten salt, where they could be held in semi-solid state, and equilibrated in the path of synchrotron radiation. This permitted for the first time the direct measurement in situ of the equilibrium shape, which could then be used to extract the surface energy anisotropy. Using the anisotropy parameters determined in these experiments, a series of phase-field simulations showed that at low supersaturation, the basal plane dendrites grow much faster than the c-axis dendrites. However, as the supersaturation increases, the growth rate for c-axis dendrites increases much more rapidly than that of the basal plane dendrites. This is due in part to the tip-splitting that occurs for the basal plane dendrites at higher supersaturations. We believe that this is due to the inability of the dendrite tip to reduce its curvature beyond a certain point, because of the forbidden orientations. These observations were surprising at first, but once we knew what to look for, we found that they are consistent with experimental results, and we are in the process of correlating these data into an article for submission to Acta Materialia.
Bibliography Type:	Description: (Last Updated: 04/03/2014)